Uncertainty-Guided Curriculum Learning for Automated Liver Fibrosis Staging on Heterogeneous MRI

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Abstract. Liver fibrosis, a major global health issue induced by chronic liver diseases, requires accurate staging for prognosis, treatment planning, and prevention of cirrhosis progression. Automated staging using MRI has emerged as a promising solution, but heterogeneity across multi-center and multi-phase MRI data—caused by variations in imaging protocols, device types, and image quality—significantly limits the generalizability and clinical transferability of deep learning models. Given such distributional discrepancies, a single model struggles to adequately capture all key features. To address this challenge, we propose an uncertainty-guided curriculum learning framework that progressively incorporates samples with higher uncertainty, thereby improving robustness in fibrosis staging. Experimental results demonstrate superior modeling of hard cases and enhanced cross-modal generalization in both cirrhosis and substantial fibrosis detection tasks. Our framework offers a clinically viable pathway to improving automated fibrosis staging on heterogeneous MRI data. The code is available at https://github.com/pazjin/FibUCL.

Keywords: Liver fibrosis staging · MRI heterogeneity · Uncertainty estimation · Domain generalization · Deep learning · Medical image analysis · Cross-modal robustness

1 Introduction

Liver fibrosis, a common outcome of chronic liver disease, can progress to cirrhosis or hepatocellular carcinoma (HCC), making accurate staging essential. Histopathology, the clinical gold standard, is invasive and limited for early screening. MRI provides non-invasive, multi-modal assessment with strong staging potential [18].

Traditional interpretation relies on radiologists' judgment, while deep learning, despite recent advances [3, 16], faces challenges with multi-modal MRI due to inter-center heterogeneity and modality redundancy or conflict.

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We propose an **uncertainty-guided progressive learning framework**. Slice-level prediction entropy identifies challenging samples for progressive training, improving robustness and generalization. Domain shifts are mitigated via Maximum Mean Discrepancy (MMD), and an attention-gated Mixture-of-Experts (MOE) enables adaptive modality fusion.
Contributions:

- Entropy-based curriculum for progressive incorporation of high-uncertainty samples.
- Attention-gated MOE with MMD for robust cross-modal and cross-center learning.
- Extensive multi-center, multi-modal validation achieving state-of-the-art fibrosis staging.

2 Related Work

Uncertainty modeling (MC Dropout, ensembles, entropy-based sampling [6,14,15,19,25]) improves robustness and hard-sample handling, yet cross-center overconfidence remains a challenge [10]. Curriculum learning and hard-sample mining [2,7,8,11,17,23] enhance convergence, but fixed-threshold approaches lack flexibility. Domain shifts are typically addressed via MMD, CORAL, or adversarial alignment [4,12,20], and multi-modal fusion (attention, gated fusion, MOE [1,9,21]) exploits complementary information. Our method unifies entropy-guided curriculum, MMD alignment, and MOE-based fusion to robustly handle cross-modal heterogeneity and challenging samples.

3 Methodology

3.1 Problem Formulation

Let the multi-modal MRI slice set be denoted as $\mathcal{X} = \{x_i^m\}_{i=1,m\in\mathcal{M}}^N$, where N represents the total number of slices and $\mathcal{M}$ denotes the set of modalities. We define a discriminative mapping function $f_\theta : \mathcal{X} \to \{0,1\}$, aiming to solve the following two binary sub-tasks simultaneously:

$$y_i^{(1)} = \begin{cases} 1, & \text{if stage is S4} \\ 0, & \text{if stage is S1–S3} \end{cases} \qquad y_i^{(2)} = \begin{cases} 1, & \text{if stage is S2–S4} \\ 0, & \text{if stage is S1} \end{cases} \tag{1}$$

Under the generalization constraints of cross-modality $\mathcal{M}$ and cross-domain $\mathcal{D}$, the overall optimization objective is formulated as:

$$\min_\theta \; \mathbb{E}_{x\sim\mathcal{D}_{\text{train}}} \, \ell\big(f_\theta(x), y\big) \; + \; \lambda \cdot \text{Dist}\big(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}\big), \tag{2}$$

where $\ell(\cdot)$ denotes the classification loss, $\text{Dist}(\cdot,\cdot)$ measures the cross-domain feature distribution discrepancy, and λ is a balancing coefficient.

3.2 Entropy-Guided Progressive Curriculum

We quantify sample-level uncertainty using predictive entropy:

$$u_i = -\sum_{c\in\{0,1\}} \hat{p}_i^{(c)} \log \hat{p}_i^{(c)}, \tag{3}$$

where higher u_i indicates greater uncertainty.

Sample Stratification. Entropy values $\{u_i\}$ are clustered (K=3) in a one-dimensional space to adaptively partition samples without manual thresholds. The clusters are ranked by average entropy, yielding three cohorts:

- **Reliably-Predicted Cohort (RPC)**: low-entropy cluster, stable predictions with high confidence;
- **Ambiguously-Predicted Cohort (APC)**: medium-entropy cluster, mostly correct but with noticeable uncertainty;
- **Challenging-Predicted Cohort (CPC)**: high-entropy cluster, misclassified or highly uncertain samples representing the most challenging cases.

Progressive Training. The ordering of cohorts follows principles of curriculum learning, where tasks of intermediate difficulty are introduced prior to the easiest or hardest cases to stabilize optimization. Specifically, beginning with the **APC** cohort provides sufficiently informative yet tractable samples, enabling the model to establish discriminative decision boundaries without being overwhelmed by noise or trivial patterns. Once a reliable representation is formed, exposure to the **CPC** cohort introduces the most challenging samples, encouraging adaptation to rare but critical failure modes. Finally, training on the **RPC** cohort consolidates confident predictions, serving as a regularization stage that prevents overfitting to high-entropy cases and reinforces overall robustness. This APC $\rightarrow$ CPC $\rightarrow$ RPC progression balances stability, adaptability, and generalization, and was further corroborated by empirical observations that alternative orders led to unstable convergence or degraded accuracy.

3.3 Multi-Modal Fusion & Domain Alignment

To address modality heterogeneity and multi-center domain shifts, we employ a Mixture-of-Experts (MOE) based dynamic fusion strategy. A gating network $G(\cdot)$ generates modality weights $\alpha_m(x)$ as:

$$\hat{y} = \sum_{m\in\mathcal{M}} \alpha_m(x)\cdot E_m(x^m), \quad \alpha_m(x) = \frac{\exp(G_m(x))}{\sum_{m'}\exp(G_{m'}(x))}. \tag{4}$$

Meanwhile, cross-domain feature alignment is achieved via Maximum Mean Discrepancy (MMD):

$$\mathrm{MMD}^2(Z_s, Z_t) = \left\| \frac{1}{N_s}\sum_{i=1}^{N_s}\phi(z_s^i) - \frac{1}{N_t}\sum_{j=1}^{N_t}\phi(z_t^j) \right\|_{\mathcal{H}}^2. \tag{5}$$

As shown in Fig. 1, MOE enables dynamic cross-modality fusion, while MMD facilitates feature consistency across centers.

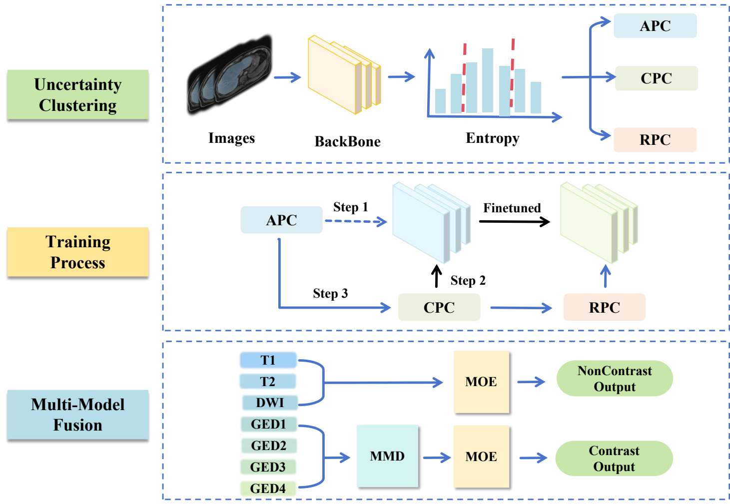

Fig. 1. Entropy-guided curriculum partitioning and multi-modal fusion architecture. **Uncertainty Clustering and Training Process:** Samples are partitioned into RPC, APC, and CPC subsets via predictive entropy, forming a progressive training pathway. **Multi-Modal Fusion:** A mixture-of-experts (MOE) framework adaptively weights modalities, while MMD loss enforces cross-domain feature alignment.

3.4 UPMA Loss: Uncertainty-Guided Progressive Multi-Domain Alignment

By integrating classification objectives with cross-domain alignment, we propose an uncertainty-guided composite loss:

$$\mathcal{L}_{\text{UPMA}} = \frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} \ell_{\text{CE}}(\hat{y}_i, y_i) + \lambda_{\text{MMD}} \cdot \text{MMD}^2(Z_s, Z_t), \tag{6}$$

where $\mathcal{S}$ denotes the current training subset, ℓ_{CE} is the cross-entropy loss, and λ_{MMD} controls the strength of domain alignment. The training proceeds in two stages: (1) $\mathcal{S} = \text{APC}$, focusing on moderately difficult cases; (2) $\mathcal{S} = \text{APC}\cup\text{CPC}$, enhancing robustness and generalization.

4 Experiments

4.1 Dataset

We constructed a multi-center liver MRI cohort of **360 patients** spanning fibrosis stages S1–S4, selected from a larger pool of 610 cases acquired on scanners from Philips and Siemens. The dataset covers **non-contrast** (T1, T2, DWI_800) and **contrast-enhanced dynamic** phases (arterial, portal, delayed, hepatobiliary) from Gd-EOB-DTPA MRI. Each case is stored in Nifty format, with occasional missing phases and without prior spatial registration [5,13,24]. Preprocessing employed `TotalSegmentator` for liver ROI extraction and slice generation. Slice-level features were extracted by modality-specific ResNet18s and fused via an attention-gated Mixture-of-Experts (MOE) with TENT-based domain adaptation.

4.2 Training Setup

In this study, ResNet18 was employed as the backbone feature extractor to perform slice-wise classification across different modalities (T1WI, T2WI, DWI_800, GED1–GED4). For non-contrast learning tasks (NonContrast), Subtask1_prob_S4 and Subtask2_prob_S1 utilized T1WI, T2WI, and DWI_800 modalities, whereas contrast learning tasks (Contrast) utilized GED1–GED4 modalities.

Training hyperparameters for samples of different entropy categories are summarized in Table 1. All experiments used a batch size of 128. Specifically, all cohort (ALL) were trained with a learning rate of 1×10^{-6} for 100 epochs; APC were trained with a learning rate ranging from 1×10^{-6} to 1×10^{-5} for 100 epochs; APC+CPC were trained with a learning rate of 1×10^{-5} for 50 epochs.

For contrast-enhanced tasks, multi-modal domain generalization using MMD was applied to mitigate distributional differences across scanning sequences: the total loss for liver cirrhosis detection was defined as $\mathcal{L}_{total} = \mathcal{L}_{cls} + 0.1 \times \mathcal{L}_{MMD}$, and for substantial fibrosis detection as $\mathcal{L}_{total} = \mathcal{L}_{cls} + 0.05 \times \mathcal{L}_{MMD}$.

Finally, multi-modal predictions were aggregated using a mixture-of-experts (MOE) framework based on an attention gating network (AttentionGatingNet). MOE was trained using the Adam optimizer with a learning rate of $1e - 3$ and binary cross-entropy loss (BCELoss) for 100 epochs, with validation performed every 20 epochs. The dataset was stratified by labels with an 80%/20% train/validation split. The gating network outputs were used to dynamically weight the contributions of each modality expert, enabling adaptive modeling according to sample difficulty.

Table 1. Training hyperparameters for different entropy categories.

Entropy Category	Learning Rate	Batch Size	Epochs
ALL	1×10^{-6}	128	100
APC	$1 \times 10^{-6} \sim 1 \times 10^{-5}$	128	100
APC + CPC	1×10^{-5}	128	50

4.3 Quantitative Results and Analysis

Contrast tasks (Tables 2, 3) showed Std outperforming early, while entropy-guided learning dominated after CPC integration, reaching near-saturation (AUC 0.980–0.998, ACC >0.955). Non-contrast tasks (Tables 4, 5) achieved similar trends: Std nearly perfected T1/T2, while entropy substantially boosted DWI_800 (0.906→0.995).

Table 2. Subtask1_prob_S4 across Contrast sequences

Contrast	**GED1**		**GED2**		**GED3**		**GED4**	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
base	0.954	0.899	0.968	0.913	0.915	0.856	0.954	0.892
APC_entropy	0.952	0.940	0.929	0.938	0.902	0.908	0.952	0.961
APC_std	0.992	**0.983**	**0.995**	**0.968**	**0.999**	**0.989**	**0.999**	0.983
APC+CPC_entropy	**0.996**	0.975	**0.995**	0.961	0.996	0.969	0.998	**0.986**
APC+CPC_std	0.995	0.966	0.994	0.961	0.994	0.960	0.998	0.980
all_entropy	0.994	0.964	0.993	0.964	0.996	0.966	0.998	0.983
all_std	0.995	0.969	0.993	0.961	0.980	0.955	0.990	0.972

Table 3. Subtask2_prob_S1 across Contrast sequences

Contrast	**GED1**		**GED2**		**GED3**		**GED4**	
	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC
base	0.841	0.835	0.872	0.821	0.874	0.808	0.906	0.833
APC_entropy	0.974	0.946	0.959	0.956	0.962	0.911	0.977	0.936
APC_std	0.985	0.954	**0.992**	**0.975**	**0.996**	**0.976**	0.994	**0.982**
APC+CPC_entropy	**0.998**	0.972	**0.992**	0.933	**0.996**	0.960	**0.998**	0.967
APC+CPC_std	0.993	0.974	0.990	0.950	0.993	0.953	0.994	0.971
all_entropy	**0.998**	**0.980**	**0.992**	0.933	**0.996**	0.966	**0.998**	0.969
all_std	0.963	0.958	0.966	0.952	0.982	0.972	0.980	0.967

4.4 External Validation and Ablation Study

We evaluated generalization on the competition validation set, comparing Adabn_Tent and MOE variants with different MMD weights (Fig. 2). Fig. 3 further compares MOE with mean-voting (MIN) across subtasks and modalities.

For *Contrast* sequences, MMD-0.1+MOE achieved the highest Subtask1 AUC (0.8050), while Subtask2 peaked with MMD-0.05+MOE (0.7822), highlighting task-dependent complementarity of domain alignment. In *NonContrast* sequences, standalone MOE reached the highest Subtask2 AUC (0.7541), suggesting limited benefit of MMD regularization under low domain shift. Across all settings, MOE consistently outperformed MIN and Adabn_Tent [22], demonstrating that uncertainty-driven progressive fusion combined with task- and modality-aware domain alignment yields robust and generalizable performance in multi-modal, multi-task fibrosis staging.

Table 4. Subtask1_prob_S4 across Non-Contrast Sequences

NonContrast	**T1**		**T2**		**DWI_800**	
	AUC	ACC	AUC	ACC	AUC	ACC
base	0.966	0.922	0.955	0.888	0.906	0.845
APC_entropy	0.937	0.936	0.961	0.949	0.970	0.903
APC_std	**0.997**	**0.977**	**1.000**	**0.992**	0.963	0.921
APC+CPC_entropy	0.994	0.961	0.998	0.978	0.995	**0.979**
APC+CPC_std	0.995	0.966	0.998	0.975	**0.997**	0.973
all_entropy	0.995	0.964	0.998	0.980	0.995	0.976
all_std	0.992	0.964	0.998	0.980	**0.996**	0.970

Table 5. Subtask2_prob_S1 across Non-Contrast Sequences

NonContrast	**T1**		**T2**		**DWI_800**	
	AUC	ACC	AUC	ACC	AUC	ACC
base	0.823	0.816	0.921	0.905	0.890	0.818
APC_entropy	0.979	0.968	0.962	0.964	0.957	0.796
APC_std	0.993	0.976	**1.000**	**0.997**	0.985	0.980
APC+CPC_entropy	**0.997**	**0.978**	0.997	0.983	**0.998**	**0.985**
APC+CPC_std	0.990	0.962	0.992	0.941	0.974	0.949
all_entropy	**0.997**	0.972	0.997	0.986	**0.998**	0.979
all_std	0.963	0.927	0.979	0.955	0.957	0.948

4.5 Performance on In-Distribution and Out-of-Distribution Test Sets

On the test set (see Table 6 for Subtask1 and Table 7 for Subtask2), *in-distribution (ID)* performance was moderate for both **Subtask1** and **Subtask2** across modalities (ACC 47.5–65.8%, AUC 67–79%). *Out-of-distribution (OOD)* results showed marked drops in AUC, especially for contrast sequences (**Subtask1** AUC 64.1%), while ACC could be misleadingly high (**Subtask2** ACC 92.9%). These results highlight the impact of domain shift and the need for robust generalization across unseen vendors.

Table 6. Test set performance (ACC and AUC) on **Subtask1** (S1–3 vs S4) across Contrast and Non-Contrast modalities. Results are reported on *In-Distribution* (Vendors A, B1, B2) and *Out-of-Distribution* (Vendor C) sets.

Modality	**Setting**	**ACC (%)**	**AUC (%)**
Contrast	ID (Vendors A, B1, B2)	47.50	78.03
	OOD (Vendor C)	32.86	64.11
Non-Contrast	ID (Vendors A, B1, B2)	47.50	79.78
	OOD (Vendor C)	32.86	46.25

Table 7. Test set performance (ACC and AUC) on **Subtask2** (S1 vs S2–4) across Contrast and Non-Contrast modalities. Results are reported on *In-Distribution* (Vendors A, B1, B2) and *Out-of-Distribution* (Vendor C) sets.

Modality	**Setting**	**ACC (%)**	**AUC (%)**
Contrast	ID (Vendors A, B1, B2)	65.83	78.20
	OOD (Vendor C)	92.86	52.62
Non-Contrast	ID (Vendors A, B1, B2)	65.83	67.27
	OOD (Vendor C)	92.86	40.31

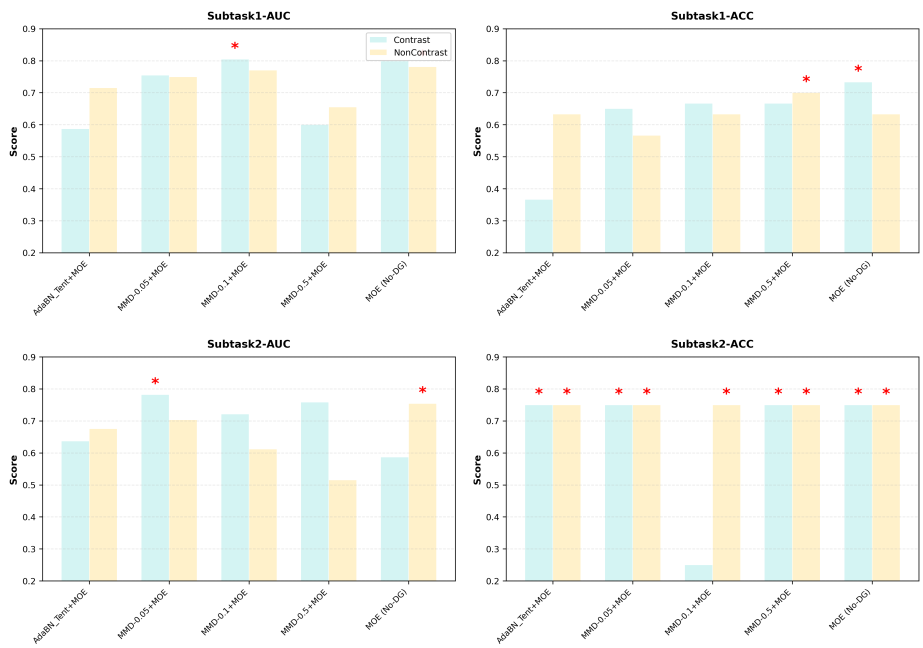

Fig. 2. Validation set performance on the competition platform across two subtasks (Contrast vs NonContrast). Best values are highlighted in *.

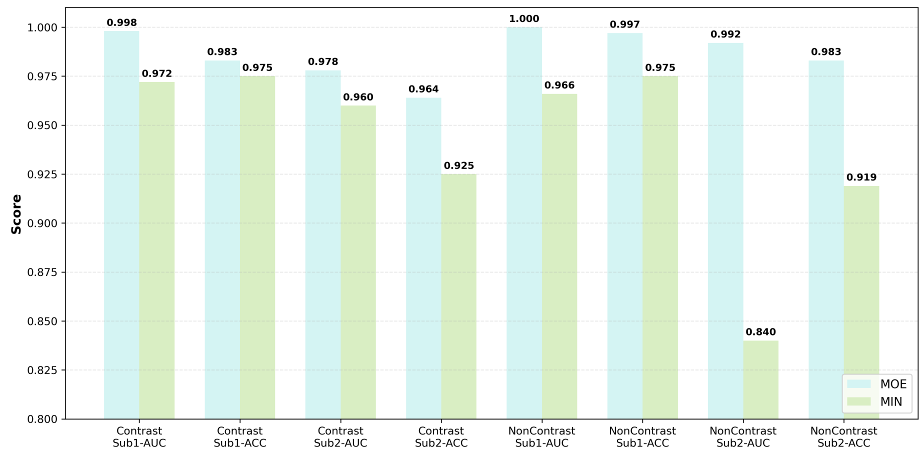

Fig. 3. Comparison of MOE and mean-voting (MIN) ensemble on different subtasks.

5 Conclusion and Future Work

We introduced an entropy-driven curriculum for MRI-based liver fibrosis staging. Progressive integration of high-uncertainty samples, combined with MMD alignment and MOE fusion, achieved moderate in-distribution (ID) performance and highlighted the impact of domain shift on out-of-distribution (OOD) test sets. Specifically, ACC ranged from 47.5% to 65.8% and AUC from 67% to 79% on ID data, while OOD results showed marked drops in AUC for contrast sequences (Subtask1 AUC 64.1%) and inflated ACC for Subtask2 (92.9%), demonstrating both the promise and limitations of our approach across unseen vendors.

These observations underscore the need for robust domain generalization, as well as careful interpretation of ACC and AUC metrics in multi-vendor settings. While the progressive curriculum stabilizes training and enables fusion across contrast and non-contrast modalities, OOD sensitivity indicates room for improvement in vendor-agnostic performance.

Future work will focus on further mitigating domain shift through advanced alignment and self-supervised strategies, evaluating models on larger multi-center datasets, and exploring computationally efficient architectures that retain the benefits of progressive fusion and uncertainty-aware learning. This direction aims to enhance robustness, interpretability, and generalization of automated multi-modal liver fibrosis staging.

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Disclosure of Interests

The authors have no competing interests to declare that are relevant to the content of this article.

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